Noisy Euclidean Correlators Do Not Certify a Spectral Gap Without Visibility

Hidden atoms, terminal moments, and multichannel witnesses

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Abstract

Euclidean correlation matrices are routinely converted into spectral gaps by generalized eigenvalue, Prony, or Lanczos procedures. We isolate a prior question: when is a positive gap identifiable from a finite correlator window? Let T be a positive self-adjoint contraction and $B_n = V^*T^nV$ its matrix-valued transfer moments. The block Hankel pencil $H_N = [B_{i+j}]$, $G_N = [B_{i+j+1}]$ gives monotone Ritz lower bounds on the visible transfer edge. Exact rank stabilization is terminal: it makes the Krylov space invariant and recovers the entire visible spectrum after finitely many moments. Outside that singular regime we prove a finite-noise no-go. For every normalized moment sequence and every $\varepsilon > 0$, mixing weight $w \le \varepsilon$ with an atom at transfer eigenvalue one changes every moment by at most w while collapsing the mass lower bound to zero. Thus no method can uniformly certify a positive gap from finite moments with nonzero error without a quantitative visibility assumption. We then give a constructive converse. If spectral mass at least γ lies at least δ beyond a claimed edge θ, a degree-N Chebyshev polynomial forces a negative Hausdorff localizer once

$$\cosh^2(N \ \mathrm{arcosh}(1 + 2\delta/\theta)) > \theta/(\delta\gamma),$$

with an explicit correction for componentwise moment error. The filter is minimax-optimal among degree-N polynomials normalized at the resolved edge. An interacting ANNNI-chain experiment up to 2^{16} states exhibits the operational content: a parity-even probe is exactly blind to the lowest odd excitation, while a two-channel (X, Z) family refutes a 35% overstated gap at Krylov degree one and recovers the transfer edge to 9.9×10^{-5} at degree six. Exact rational certificates, the full numerical pipeline, a Colab notebook, and a Lean-checked hidden-atom core are linked to fixed repository artifacts.

1 The identifiability question

A spectral gap is a support exclusion, not merely a fitted exponential slope. In a transfer formulation, after removing the vacuum, it is the assertion

$$\mathrm{spec}(T) \subset [0, e^{-m}] \qquad (m > 0), \tag{1}$$

for a positive self-adjoint contraction T. Numerical and lattice calculations generally do not access T directly. They access finitely many matrices

$$B_n = V^*T^nV, \qquad n = 0, 1, \ldots, M, \tag{2}$$

where the columns of $V : \mathbb{C}^d \to \mathcal{K}$ are states created by a chosen family of interpolating operators. These are Hausdorff moments of a positive matrix-valued spectral measure. This observation underlies generalized eigenvalue methods, Prony-type reconstruction, Rayleigh–Ritz and block Lanczos spectroscopy [1, 2, 3, 4, 5].

The inverse problem has two logically different targets:

(i) estimate spectral values that are visible to the chosen probes;

(ii) certify that no spectral value relevant to the claimed gap has been missed.

The first is an approximation problem. The second is an identifiability problem. A highly accurate estimate does not answer the second question if a small spectral component can hide below the error floor.

This paper gives a three-branch answer.

Exact terminal branch. Rank stabilization of a block Hankel matrix is a finite, checkable condition under which the Krylov approximation terminates and the visible spectrum is recovered exactly.

No-go branch. For every positive componentwise error radius, a spectral atom at transfer eigenvalue one can be hidden inside the data ball. No algorithm can infer a uniform positive gap over the unconstrained model class.

Visibility branch. A lower bound on the mass of any missed edge component turns the problem back into a finite certificate. A Chebyshev filter yields an explicit depth and noise margin.

The algebraic ingredients are classical: positive moment problems, Krylov termination, and Chebyshev acceleration. The claim here is the assembled operational boundary—including the impossibility theorem, the minimal extra visibility datum, and an auditable correlator-only witness—not the invention of Rayleigh–Ritz or of the truncated moment problem.

2 Matrix moments and the visible spectral edge

Let $\mathcal{K}$ be a complex Hilbert space, let $0 \le T \le \mathbf{1}$ be self-adjoint, and let $V : \mathbb{C}^d \to \mathcal{K}$ be bounded. Write E_T for the spectral resolution of T and

$$\Sigma(A) = V^* E_T(A) V, \qquad B_n = \int_0^1 x^n \, d\Sigma(x) = V^* T^n V. \tag{3}$$

The visible cyclic subspace and its edge are

$$\mathcal{K}_V = \overline{\operatorname{span}}\{T^j \operatorname{ran} V : j \ge 0\}, \qquad \rho_V = \|T|_{\mathcal{K}_V}\|. \tag{4}$$

Only $T|_{\mathcal{K}_V}$ is identifiable from the moments. If $\mathcal{K}_V \ne \mathcal{K}$, a single probe family is blind to the orthogonal reducing complement.

For $N \ge 0$ define block Hankel matrices

$$H_N = [B_{i+j}]_{i,j=0}^N, qquad G_N = [B_{i+j+1}]_{i,j=0}^N. \tag{5}$$

Given $c = (c_0, \dots, c_N) \in (\mathbb{C}^d)^{N+1}$, set

$$W_N c = \sum_{j=0}^N T^j V c_j, \qquad q_c(x) = \sum_{j=0}^N x^j c_j. \tag{6}$$

Then

$$\begin{aligned} c^* H_N c &= \|W_N c\|^2, \\ c^* G_N c &= \langle W_N c, T W_N c \rangle, \\ c^*(\theta H_N - G_N) c &= \int_0^1 (\theta - x) \, q_c(x)^* d\Sigma(x) q_c(x). \end{aligned} \tag{7}$$

Proposition 2.1 (Ritz/localizer dictionary). *Let*

$$\rho_N = \sup_{c \notin \ker H_N} \frac{c^* G_N c}{c^* H_N c}. \tag{8}$$

Then ρ_N is the largest Ritz value of T on $\mathcal{K}_N = \operatorname{ran} W_N$, and

$$0 \le \rho_N \le \rho_{N+1} \le \rho_V, \qquad \rho_N \uparrow \rho_V. \tag{9}$$

Moreover, $\theta H_N - G_N \not\succeq 0$ is a rigorous finite falsifier of $\rho_V \le \theta$, whereas $\theta H_N - G_N \succeq 0$ at one finite N is not in general a certificate of that support bound.

Proof. The first two identities in Eq. (7) identify the generalized Rayleigh quotient with that of T on $\mathcal{K}_N$. The inclusions $\mathcal{K}_N \subseteq \mathcal{K}_{N+1}$ give monotonicity. Their union is dense in $\mathcal{K}_V$, so the variational suprema converge to $\|T|_{\mathcal{K}_V}\|$. If a localizer has a negative direction, Eq. (7) is incompatible with spectral support in $[0, \theta]$. Positivity on polynomials of bounded degree does not exclude spectral weight detectable only at larger degree. □

This proposition separates a common ambiguity. A Ritz value is a lower bound on the visible transfer edge and hence an *upper* bound on the visible mass $-\log \rho_V$. A negative localizer falsifies an overstated mass. Neither becomes a full-gap theorem until probe completeness is supplied.

3 The exact terminal branch

Finite exact recovery occurs when the moment problem becomes degenerate. This is the operator form of flat propagation in truncated moment theory and of exact termination in block Lanczos [8, 9, 10, 4].

Theorem 3.1 (Flat Krylov termination). *If*

$$\operatorname{rank} H_N = \operatorname{rank} H_{N+1} < \infty, \tag{10}$$

then $\mathcal{K}_N = \mathcal{K}_{N+1}$, the space $\mathcal{K}_N$ is invariant under T, and

$$\mathcal{K}_V = \mathcal{K}_N, qquad \rho_V = \rho_N. \tag{11}$$

The finite generalized pencil (G_N, H_N) on the quotient by $\ker H_N$ recovers the complete visible spectrum, with multiplicities in the cyclic representation. If in addition $\mathcal{K}_V = \mathcal{K}$ and $T = e^{-aH}$ on the excitation space, the recovered edge certifies the full gap $m = -a^{-1} \log \rho_N$ (with the usual extended-value convention at $\rho_N = 0$).

Proof. Since $H_N = W_N^* W_N$, $\operatorname{rank} H_N = \dim \mathcal{K}_N$. The nesting $\mathcal{K}_N \subseteq \mathcal{K}_{N+1}$ and Eq. (10) therefore imply equality. But $T\mathcal{K}_N \subseteq \mathcal{K}_{N+1}$, hence $T\mathcal{K}_N \subseteq \mathcal{K}_N$. All further Krylov spaces equal $\mathcal{K}_N$, so their closure is $\mathcal{K}_V = \mathcal{K}_N$. The compressed operator is now the exact restriction rather than an approximation, and the quotient pencil represents it in the nonorthogonal Krylov coordinates. □

Example 3.2 (Exact rational terminal certificate). Take

$$T = \operatorname{diag}(1/5, 1/2, 4/5), \qquad V = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

The tracked exact-arithmetic artifact constructs $B_0, \ldots, B_5$ and finds

$$\operatorname{rank} H_0 = 2, \qquad \operatorname{rank} H_1 = \operatorname{rank} H_2 = 3.$$

On a rational quotient basis the pencil determinant is

$$-\frac{9}{5000}(2\lambda - 1)(5\lambda - 4)(5\lambda - 1), \tag{12}$$

recovering $\{1/5, 1/2, 4/5\}$ exactly and hence the visible mass $-\log(4/5) = \log(5/4)$. No floating-point rank decision enters this example.

Flatness is powerful but singular. With noisy data, exact rank is destroyed; thresholding small singular values introduces a model choice. The next section shows that this is not merely a numerical inconvenience.

4 A finite-noise no-go theorem

The obstruction already appears for a scalar normalized spectral measure. Its matrix analogue follows by adding an identity-valued atom.

Theorem 4.1 (Hidden-atom indistinguishability)**.** *Let μ be a probability measure on $[0,1]$ with moments $m_n = \int x^n \, d\mu(x)$. For $0 < w \le 1$ define*

$$\mu_w = (1-w)\mu + w\delta_1, \qquad m_n^{(w)} = (1-w)m_n + w. \tag{13}$$

Then, for every $n \ge 0$,

$$|m_n^{(w)} - m_n| = w(1 - m_n) \le w. \tag{14}$$

The measure μ_w has transfer edge one and therefore zero mass lower bound. Consequently, for every finite window and every componentwise error radius $\varepsilon > 0$, the error ball around the moments of a gapped measure contains the moments of a zero-gap measure.

If Σ is an I_d-normalized positive matrix measure and $B_n = \int x^n d\Sigma(x)$, the same conclusion holds in operator norm for

$$\Sigma_w = (1-w)\Sigma + wI_d\delta_1, \qquad B_n^{(w)} = (1-w)B_n + wI_d, \tag{15}$$

because $0 \preceq B_n \preceq I_d$ and $\|B_n^{(w)} - B_n\| \le w$.

Proof. Since $0 \le x^n \le 1$ on $[0,1]$, one has $0 \le m_n \le 1$. Equation (14) follows by direct subtraction. The added atom places positive spectral mass at $x = 1$, so the support edge is one and $-\log 1 = 0$. Choose any $0 < w \le \min\{\varepsilon, 1\}$. The matrix statement is identical after replacing scalar order by Loewner order. $\square$

Corollary 4.2 (Algorithm-independent impossibility)**.** *No estimator whose only input is a finite positive-moment window with a nonzero componentwise error bar can return a uniformly valid strictly positive gap lower bound over all positive measures on $[0,1]$.*

This conclusion applies equally to nonlinear fits, Bayesian reconstructions, GEVP, Prony, Lanczos, Padé, semidefinite programs, and machine-learning estimators. It is a property of the information set, not of the solver. A denoising projection onto the positive Hankel cone can improve consistency and variance [7]; it cannot exclude the hidden measure in Theorem 4.1 without adding a visibility prior.

The theorem is deliberately worst-case. It does not say that small atoms are common or that gap estimation is useless. It says that a rigorous positive certificate must disclose what rules them out.

5 Visibility restores a finite certificate

We now add exactly the missing quantitative datum. Work first with a scalar probability measure ν on $[0,1]$. Fix a claimed edge $0 < \theta < 1$, a resolution $0 < \delta \le 1-\theta$, and a visibility floor $0 < \gamma \le 1$.

Definition 5.1 (Visibility alternative)**.** The pair (δ, γ) is visible beyond θ if every admissible measure whose edge reaches $\theta + \delta$ satisfies

$$\nu([\theta + \delta, 1]) \ge \gamma. \tag{16}$$

The assumption is falsifiable and model-dependent. It can come from a frame bound for an operator family, a symmetry-resolved overlap estimate, a source theorem, or an experimentally calibrated minimum spectral weight. It cannot be manufactured from the same noisy moments without circularity, by Theorem 4.1.

Let T_N denote the degree-N Chebyshev polynomial of the first kind and define

$$p_N(x) = \frac{\mathsf{T}_N(2x/\theta - 1)}{\mathsf{T}_N(1 + 2\delta/\theta)}, \qquad \alpha_N = \frac{1}{\mathsf{T}_N(1 + 2\delta/\theta)}. \tag{17}$$

Theorem 5.2 (Visibility-conditioned Chebyshev witness)**.** *If Eq. (16) holds, then the degree-N localizer value*

$$Q_N = \int_0^1 (\theta - x) p_N(x)^2 \, d\nu(x) \tag{18}$$

satisfies

$$Q_N \le \theta \alpha_N^2 - \delta\gamma. \tag{19}$$

In particular $Q_N < 0$, and the claimed support bound is falsified, whenever

$$\cosh^2(N \operatorname{arcosh}(1 + 2\delta/\theta)) > \frac{\theta}{\delta\gamma}. \tag{20}$$

If $\delta\gamma < \theta$, it is sufficient that

$$N > \frac{\operatorname{arcosh}\sqrt{\theta/(\delta\gamma)}}{\operatorname{arcosh}(1 + 2\delta/\theta)}. \tag{21}$$

Proof. On $[0,\theta]$, the affine argument of T_N lies in $[-1,1]$, so $|p_N| \le \alpha_N$. On $[\theta+\delta, 1]$, the argument is at least $1 + 2\delta/\theta$; monotonicity of T_N on $[1,\infty)$ gives $p_N \ge 1$. Split Eq. (18) into $[0,\theta]$, $(\theta, \theta+\delta)$, and $[\theta+\delta, 1]$. The three contributions are at most $\theta\alpha_N^2$, 0, and $-\delta\gamma$, respectively. Finally use $\mathsf{T}_N(y) = \cosh(N \operatorname{arcosh} y)$ for $y \ge 1$. □

Proposition 5.3 (Minimax stop-band leakage)**.** *Among all real polynomials r of degree at most N satisfying $r(\theta + \delta) = 1$, the smallest possible uniform leakage on the claimed support is*

$$\inf_{\substack{\deg r \le N \\ r(\theta+\delta)=1}} \max_{0 \le x \le \theta} |r(x)| = \frac{1}{\mathsf{T}_N(1 + 2\delta/\theta)} = \alpha_N. \tag{22}$$

The filter p_N in Eq. (17) attains the infimum.

Proof. Under $y = 2x/\theta - 1$, the interval $[0,\theta]$ becomes $[-1,1]$ and the normalization point becomes $y_0 = 1 + 2\delta/\theta > 1$. The Chebyshev extremal property states that a degree-N polynomial bounded by one on $[-1,1]$ has absolute value at most $\mathsf{T}_N(y_0)$ at y_0. Rescaling any candidate r therefore gives the lower bound in Eq. (22). The numerator of p_N has sup norm one on the interval and value $\mathsf{T}_N(y_0)$ at the normalization point, so equality holds. □

Thus Theorem 5.2 uses the smallest possible uniform low-band amplitude for its chosen degree and edge normalization. This does not make Eq. (20) minimax over every conceivable nonlinear estimator or every weighted error model; it establishes optimal separation within the polynomial localizer class actually accessible from a finite moment window.

Corollary 5.4 (Matrix-valued multichannel witness)**.** *Let $\Sigma([0,1]) = I_d$ and set $\nu = d^{-1}\operatorname{tr}\Sigma$. If $\nu([\theta+\delta,1]) \geq \gamma$ and Eq. (20) holds, then*

$$\theta H_N - G_N \not\succeq 0. \tag{23}$$

Hence the matrix correlator window contains a finite direction that falsifies the claimed edge.

Proof. Let $a_0, \ldots, a_N$ be the coefficients of p_N and, for each standard basis vector $e_\ell \in \mathbb{C}^d$, take the block coefficient vector $c^{(\ell)} = (a_0 e_\ell, \ldots, a_N e_\ell)$. Then

$$\frac{1}{d}\sum_{\ell=1}^{d} (c^{(\ell)})^*(\theta H_N - G_N)c^{(\ell)} = Q_N < 0.$$

At least one summand is negative. □

The degree grows only logarithmically with $1/\gamma$ at fixed resolution. For small δ/θ, however, $\operatorname{arcosh}(1+2\delta/\theta) \sim 2\sqrt{\delta/\theta}$, exposing the expected edge-resolution cost. This is closely related to the Chebyshev mechanism in Kaniel–Paige–Saad convergence bounds, but here the output is a one-sided moment localizer with an explicit visibility premise, not merely an eigenvalue convergence estimate [11, 4].

5.1 Certified moment error

Write $p_N(x) = \sum_{j=0}^{N} a_j x^j$ and suppose measured moments satisfy $|\widetilde{m}_k - m_k| \leq \varepsilon$ for $0 \leq k \leq 2N+1$. Form the measured localizer $\widetilde{Q}_N$ by expanding Eq. (18) in those moments.

Proposition 5.5 (Noise-safe sign test)**.** *The deterministic error obeys*

$$|\widetilde{Q}_N - Q_N| \leq (1+\theta)\varepsilon\left(\sum_{j=0}^{N}|a_j|\right)^2 =: R_N. \tag{24}$$

Therefore

$$\widetilde{Q}_N + R_N < 0 \tag{25}$$

is a rigorous noisy-data falsifier of support in $[0,\theta]$. Under the visibility alternative, the sufficient worst-case condition

$$\theta\alpha_N^2 - \delta\gamma + 2R_N < 0 \tag{26}$$

guarantees detection for every admissible error realization.

Proof. If $p_N^2 = \sum_{k=0}^{2N} b_k x^k$, then $Q_N = \theta\sum b_k m_k - \sum b_k m_{k+1}$. Hence the error is at most $(1+\theta)\varepsilon\sum|b_k|$. Convolution gives $\sum|b_k| \leq (\sum|a_j|)^2$. Under the visibility alternative, $\widetilde{Q}_N + R_N \leq Q_N + 2R_N \leq \theta\alpha_N^2 - \delta\gamma + 2R_N$, which proves the stated worst-case condition. □

Power-basis coefficients can grow rapidly. Thus increasing N is not monotonically beneficial at fixed componentwise precision: spectral localization improves while the rigorous error radius may worsen. The tracked stress test with $\theta = 0.8$, $\delta = 0.1$, $\gamma = 10^{-2}$ and $\varepsilon = 10^{-14}$ becomes negative at degree five; the analytic sufficient bound turns negative at degree six. Higher precision or a stable recurrence-based error model can extend the useful window, but must be declared explicitly.

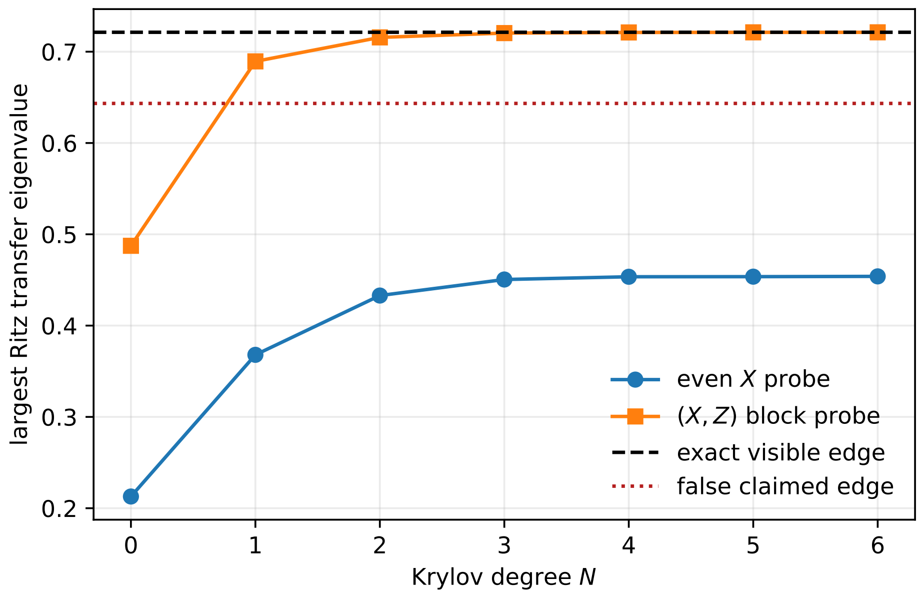

Figure 1: Largest Ritz transfer value for $L = 12$. The even X probe converges to the lowest visible even excitation and misses the true edge. The (X, Z) block family converges rapidly to the exact edge.

6 Interacting spin-chain test

Theorems about visibility should be tested in a model where blindness is exact and the transfer operator is not inserted by hand. We use the open quantum ANNNI chain

$$H_L = -J_1 \sum_{j=1}^{L-1} Z_j Z_{j+1} - J_2 \sum_{j=1}^{L-2} Z_j Z_{j+2} - h_x \sum_{j=1}^{L} X_j, \tag{27}$$

at $(J_1, J_2, h_x) = (1, 0.37, 2.2)$. This generic next-nearest-neighbour interacting instance is not treated through a free-fermion solution. It preserves the global parity $P = \prod_j X_j$. The computed ground state is even and the first excited state odd throughout $L = 6, 8, \ldots, 16$.

Let $|0\rangle$ be the independently computed ground state and, at the central site, define centered probes

$$|\psi_X\rangle = (X_c - \langle X_c\rangle)|0\rangle, \qquad |\psi_Z\rangle = (Z_c - \langle Z_c\rangle)|0\rangle. \tag{28}$$

Because X_c is parity even, its overlap with the first odd state vanishes exactly. Because Z_c is odd, it can see that state. With $\tau = 0.2$ we construct only the correlator matrices

$$B_n^{ab} = \langle\psi_a|e^{-n\tau(H_L - E_0)}|\psi_b\rangle, \qquad a, b \in \{X, Z\}. \tag{29}$$

The moment pipeline is then separated from the low-energy diagonalization used as ground truth.

Figures 1 and 2 show the two distinct outputs. The Ritz value estimates an edge; the negative localizer certifies that a proposed smaller edge is impossible for the measured spectral measure. The single even probe can look numerically stable while converging to the wrong sector.

At $L = 16$ the Hilbert-space dimension is 65,536. The independent gap is 1.5293121573, the true transfer edge is 0.7364879301, and the degree-six block value is 0.7363890221. Its absolute error is 9.89×10^{-5}. The first-state weights are $w_X = 1.17 \times 10^{-31}$ and $w_Z = 0.26437$. These numbers do not prove a thermodynamic mass gap; they validate the visibility mechanism and the correlator-only certificate at nontrivial finite size.

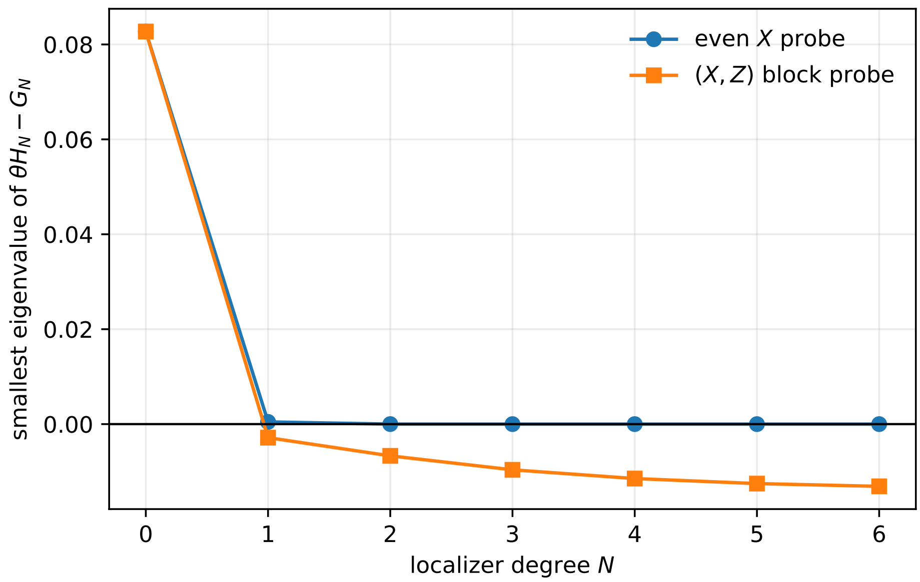

Figure 2: Smallest eigenvalue of the localizer for a gap overstated by 35%. The multichannel family gives a negative certificate from degree one; the symmetry-blind X family never does.

Table 1: ANNNI scaling. w_X, w_Z are squared overlaps with the first excited state. $\widehat{\rho}_6$ is the block-Krylov edge from moments through order 13. All rows refute the 35% overstated gap at degree one; the X-only family never refutes it through degree six.

L	$E_1 - E_0$	$\rho = e^{-.2(E_1-E_0)}$	w_X	w_Z	$\lvert\widehat{\rho}_6 - \rho\rvert$
6	2.0845863	0.6590755	$3.4\,10^{-31}$	0.45160	$1.9\,10^{-7}$
8	1.8574804	0.6897017	$4.3\,10^{-32}$	0.39733	$3.9\,10^{-6}$
10	1.7217194	0.7086852	$9.8\,10^{-34}$	0.35390	$2.1\,10^{-5}$
12	1.6335802	0.7212885	$2.0\,10^{-32}$	0.31837	$5.7\,10^{-5}$
14	1.5729200	0.7300925	$2.8\,10^{-30}$	0.28900	$8.6\,10^{-5}$
16	1.5293122	0.7364879	$1.2\,10^{-31}$	0.26437	$9.9\,10^{-5}$

7 Reproducibility and formal boundary

All computational claims are generated from public, deterministic artifacts:

- exact rational flat-extension, hidden-atom, and Chebyshev artifacts;
- sparse ANNNI construction, independent low-energy benchmark, moment construction, block pencils, and localizers;
- tracked local JSON for $L = 6, 8, 10, 12$;
- a Colab summary for $L = 12, 14, 16$, including Python/NumPy/SciPy versions, source commit, and full-result SHA-256;
- scripts that regenerate all three figures.

Running the tracked `verify_all.py` from the repository root regenerates the exact artifact and audits every theorem-level invariant quoted from the local and Colab records; its README gives the one-line command. The verification directory is publicly archived in the companion repository. The Colab entry point is `colab_annni_scale.ipynb`.

The algebraic core of Theorem 4.1 is separately formalized in Lean 4 at fixed commit fixed Lean source, with the axiom audit and draft PR #25. The build checks 8,158 targets with

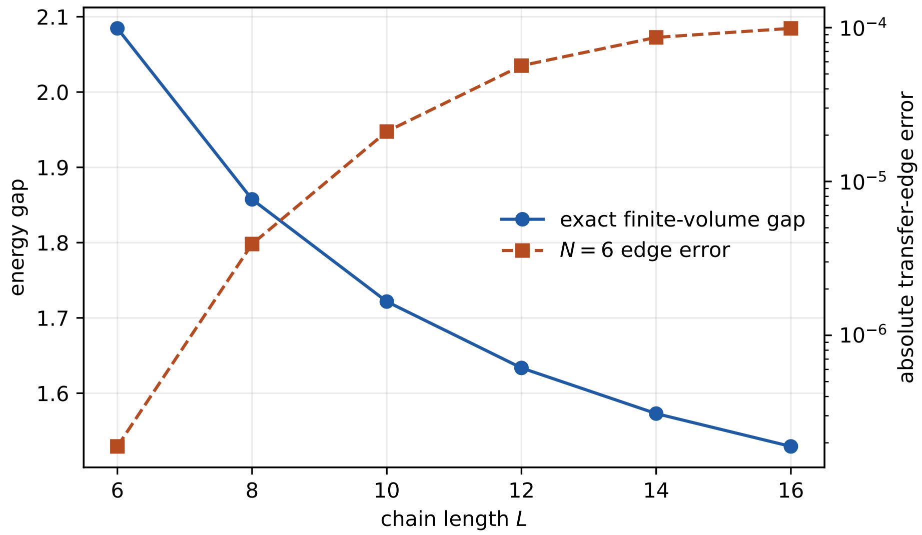

Figure 3: Independent finite-volume energy gap and degree-six transfer-edge error. The $L = 12, 14, 16$ run was executed in Colab Pro+ on the fixed source commit recorded in the artifact. This is a finite-size validation, not a thermodynamic-gap extrapolation.

no `sorry` and no project axioms. The reported theorem dependencies are exactly {`propext`, `Classical.choice`, `Quot.sound`}.

The formalization proves the normalized hidden-atom mixture, componentwise distance bound, preservation of the interval constraints, existence inside every error ball $0 \le \varepsilon \le 1$, and $-\log 1 = 0$. It does not formalize the spectral theorem, Chebyshev integral bound, or ANNNI numerics. Those remain paper proofs and reproducible computations, respectively.

8 Relation to prior methods and novelty audit

Matrix Euclidean correlators as moment data, their relation to Rayleigh–Ritz/Lanczos, and rigorous bounds on smeared spectral functions are developed in [5]. The Prony–Ritz equivalence class and exact solution of finite-dimensional spectra from sufficient correlation data are analysed in [4]; matrix-polynomial combinations of GEVP and Prony appear in [2]. GEVP convergence with increasing operator bases is classical in lattice spectroscopy [1], and direct imaginary-time gap estimators have been tested in many-body models in [6]. Recursive and flat matrix moment problems are studied in [8, 9, 10]. None of those facts is claimed as new here.

The main conceptual advance is therefore a boundary theorem: finite noisy correlators support rigorous gap falsification, and under disclosed visibility they support finite detection, but they do not support an unconditional positive gap certificate. Exact flatness is the exceptional terminal regime.

9 Consequences and limitations

For lattice mass-gap programmes. A finite correlator analysis can disprove an overstated transfer gap by a negative localizer. To prove a positive volume-uniform gap, one still needs a volume-uniform totality or visibility theorem for the chosen gauge-invariant operator family, controlled errors, and the reconstruction connecting the Euclidean measure to a positive transfer operator. The present results do not establish any of those model-dependent inputs for four-dimensional Yang–Mills.

Table 2: Claim audit. "This paper" means the stated combination and specialization, not ownership of the classical ingredients.

Item	Prior status	Contribution here
Matrix moments and block Ritz	Established in moment and lattice spectroscopy	Unified notation tied to a one-sided gap localizer
Exact finite recovery	Standard Krylov termination/flat moment phenomenon	Checkable terminal criterion stated directly for visible transfer gaps and verified by an exact rational artifact
Finite-noise hidden edge	Elementary mixture mechanism; small-component instability is familiar in inverse problems	Algorithm-independent zero-gap construction inside every positive moment error ball, with a matrix version and Lean-checked core
Minimum visibility	Minimum amplitude assumptions occur in support recovery	Identified as the missing datum for rigorous gap certification, rather than an optional numerical regularizer
Chebyshev filtering	Classical in extremal eigenvalue convergence	Explicit localizer sign theorem with $(\theta, \delta, \gamma, N)$ depth and a componentwise error radius, plus minimax optimality of its uniform stop-band leakage among normalized degree-N polynomial filters
Probe blindness	Symmetry selection rules are standard	Controlled interacting example separating a stable wrong-sector estimate from a multichannel finite certificate through 2^{16} states

For operator design. Adding channels is not merely a variance-reduction strategy. It changes the visible cyclic subspace. Symmetry-resolved operator families should be judged by quantitative frame or overlap bounds, not only by the apparent stability of their extracted levels.

For noisy moment algorithms. Projection to the positive Hankel cone is useful for restoring consistency, but a projected point estimate should not be reported as a support theorem. The visibility floor and the propagated sign margin in Eq. (25) are the appropriate certificate metadata.

Limitations. The Chebyshev stop-band factor is minimax optimal only within normalized degree-N scalar polynomial filters; the resulting mass-versus-leakage sign condition is sufficient, not a necessary identifiability threshold. The componentwise error model ignores correlations and can be conservative. The ANNNI study is finite-volume and uses exact sparse propagation rather than Monte Carlo data. No thermodynamic extrapolation is attempted. Matrix-valued optimal filters, correlated confidence regions, interval arithmetic, and formalization of the Chebyshev theorem remain open extensions.

10 Conclusion

The distinction between estimating a visible level and certifying a spectral gap is structural. Block Hankel pencils give monotone visible-edge estimates and finite falsifiers. Exact rank stabilization closes the problem at finite depth. With any nonzero error and no visibility premise, however, a hidden atom at transfer eigenvalue one lies inside the data ball and destroys every uniform positive gap conclusion. A quantitative visibility floor is the minimal bridge back to a finite theorem; Chebyshev localization makes its cost explicit. The interacting spin-chain test shows why this distinction matters: a symmetry-blind probe can converge smoothly and still miss the gap-controlling state, while a multichannel family supplies an immediate negative witness.

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